Simplified Global Analysis of Parton Distribution Functions (PDFs)

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Abstract

The purpose of this project is to perform a simplified global analysis of parton distribution functions (PDFs) using gradient descent, the finite-difference method, and the QR algorithm. As we will see, however, classical methods are insufficient for accurately extracting PDFs from data, leading to the need for more specialized algorithms like the Levenberg-Marquardt optimization scheme. At the end of this report, I will present PDFs for the u_v, $\bar{u}$, d_v, $\bar{d}$, s, and $\bar{s}$ quarks obtained from fitting to ep deep inelastic scattering (DIS) data.

Introduction

In the context of nuclear/particle physics, a parton is a constituent subatomic particle (i.e. a quark or gluon) that makes up a larger hadron (e.g. a proton or neutron), and a parton distribution function $f(x)$ is a probability distribution that describes the probability of finding a particular parton carrying a fraction x of the parent hadron's total momentum at a given energy scale. PDFs are essential for determining the internal structure of hadrons, allowing us to calculate various cross sections within a scattering experiment. Once determined, they can be applied universally to any calculation, extending the use of PDFs beyond the experiments from which they were determined.

The general process for determining PDFs via global analysis begins with amalgamating data from deep inelastic scattering (DIS) experiments, Drell-Yan processes, and jet production measurements, providing a "global" view of the different observables within particle collisions. Based on the theory of quantum chromodynamics (QCD), the observables of these experiments can be expressed in terms of PDFs, allowing us to calculate theoretical values for the observables if the PDFs are already known. Of course, we do not know the exact function for a given PDF (in fact, the goal is to *find* that function), but we can guess its form up to a set of arbitrary parameters based on physical constraints and plots of the experimental data. Once a parameterized 'ansatz' is chosen, we can calculate theoretical values for the experimental observables and then use a χ^2-minimization scheme to determine the best fit parameters for each of the PDFs we wish to find [1, 2]. In order to account for the error accumulated from all of the data sets in the analysis, we can then calculate

the uncertainty in our fitted PDFs using approximate error propagation algorithms (e.g. the Hessian matrix method [2, 3, 4]) or statistical analysis (e.g. Monte Carlo sampling of parameter space [5, 6, 7]).

To keep this project simple, I will only consider data from electron-proton and positron-proton DIS experiments performed at the same average energy scale ($Q^2 = 15$ GeV). While a true global analysis incorporates a much more diverse set of data and makes use of the DGLAP equations to interpolate between different energy scales [8, 1], such considerations are beyond the scope of this course and hence will be ignored. Instead, I wish to perform a simplified, minimal global analysis in order to highlight the computational techniques and how they fit into the general procedure for obtaining PDFs. In what follows, I will first discuss the theoretical framework behind global analysis (reduced to the scope of this project) and then outline its corresponding computational implementation. In our examination of the latter, I will take note of the algorithms that I implemented in the analysis, commenting on the limitations that I faced and the solutions I came up with in response. At the end, I will present my finalized PDF fits and briefly interpret their physical implications.

Theoretical Framework

For this project, I will assume that parton distribution functions f take on the generic form

$$f(x, Q^2) = a_0 x^{a_1} (1 - x)^{a_2} (1 + a_3 \sqrt{x}), \tag{1}$$

where x is the momentum fraction of the parton and Q^2 is the experimental energy scale that determines the parameters $\{a_i\}$. As noted previously, in order to simplify my calculations I will only consider constant Q^2. However, I would like to emphasize once more that in real global analyses, it is crucial to evolve Q^2 via the DGLAP equations in order to determine how the parameters of the PDFs change with the energy scale of an experiment.

As an aside on notation, $f(x, Q^2)$ refers to the general functional form that all of the PDFs in my analysis will follow. When I eventually refer to the PDF of a specific quark q, I will denote it as $q(x)$, where I have dropped the Q^2 dependence for brevity of notation since I will only be considering constant Q^2. It should be kept in mind, though, that $q(x) = f(x, Q^2; \{a_i\})$ for a particular set of parameters $\{a_i\}$ unique to that specific quark.

To further simplify my global analysis, I will only consider the internal structure of the proton, which is typically measured through an observable called the "F_2 structure function." In terms of the PDFs for a quark q and its corresponding antiquark $\bar{q}$, the F_2 structure function (at a fixed scale Q^2) is defined as

$$F_2(x) = \sum_i e_i^2 x \left[q_i(x) + \bar{q}_i(x) \right], \tag{2}$$

where e_i^2 is the charge of the ith quark type.

In general, a quark's distribution $q(x)$ can be decomposed into the sum of a distribution of valence quarks $q_v(x)$ (the physically "relevant" quarks that determine a hadron's quantum numbers) and a distribution of sea quarks $q_s(x)$ (the temporary quarks that are created during collisions and then immediately annihilated by their corresponding antiquarks). Assuming that sea quarks and antiquarks are created in equal amounts during a collision, we

can say that the distribution of sea quarks is equal to the distribution of antiquarks, i.e. $q_s(x) = \bar{q}(x)$, so that

$$q(x) = q_v(x) + \bar{q}(x) \tag{3}$$

Then, the F_2 structure function (eq. 2) can be written in terms of valence quark and antiquark distributions as:

$$F_2(x) = \sum_i e_i^2 x \left[q_{v_i}(x) + 2\bar{q}_i(x) \right]. \tag{4}$$

For a proton collision in a DIS experiment, the main quark contributions are up and down quarks (u and d respectively) since protons are made of two up (valence) quarks and one down (valence) quark. There are also other sea quarks created during the collision process like the charm (c) and strange (s), but for the sake of simplifying this project I will only focus on the distributions of the up, down, and strange quarks and their corresponding antiparticles. As such, the F_2 structure function that I will use in my global analysis is

$$F_2(x) = x \left[\frac{4}{9} \left(u_v(x) + 2\bar{u}(x) \right) + \frac{1}{9} \left(d_v(x) + 2\bar{d}(x) \right) + \frac{1}{9} \left(s(x) + \bar{s}(x) \right) \right], \tag{5}$$

where the charge of the up quark is $e_u = \frac{2}{3}$ and the charge of the down and strange quarks is $e_d = \frac{1}{3}$. In order to reduce the number of parameters that need to be fitted later on, it is common practice to express s and $\bar{s}$ in terms of the other four PDFs. In particular, since the strange/anti-strange quarks make up a fraction of the total sea quarks created after a collision, we can assume that their corresponding PDFs are also fractions of the sea quark PDFs, which we have assumed to be equal to $\bar{u}$ and $\bar{d}$. Because strange quarks are not primary components of the proton – that is, they are not the physically "relevant" valence quarks – we can further assume that the strange and anti-strange quarks are created in pairs that annihilate each other so that the distribution of matter strange quarks s is equivalent to the distribution of their antimatter counterpart $\bar{s}$. Hence, the simplest relation we can formulate between s and $\bar{s}$ in terms of the antimatter/sea quarks $\bar{u}$ and $\bar{d}$ is

$$s = \bar{s} = \frac{1}{2} \kappa \left(\bar{u} + \bar{d} \right), \tag{6}$$

where κ is an arbitrary scaling factor, such that eq. 5 becomes

$$F_2(x) = x \left[\frac{4}{9} \left(u_v(x) + 2\bar{u}(x) \right) + \frac{1}{9} \left(d_v(x) + 2\bar{d}(x) \right) + \frac{1}{9} \kappa \left(\bar{u}(x) + \bar{d}(x) \right) \right]. \tag{7}$$

To further constrain u_v and d_v, we can impose the following sum rules

$$n_u = \int_0^1 \mathrm{d}x \, u_v(x) = 2 \tag{8}$$

$$n_d = \int_0^1 \mathrm{d}x \, d_v(x) = 1, \tag{9}$$

which follow from the fact that a single scattered proton is made from $n_u = 2$ up quarks and $n_d = 1$ down quark. In practice, these sum rules are used to constrain the normalization

factor a_0 of the u_v and d_v PDFs. If we substitute our functional form, eq. 1, into the above sum rules, we can obtain a general equation for the normalization factor of the ith valence quark:

$$a_0 = \frac{n_i}{\int_0^1 \mathrm{d}x\, x^{a_1}(1-x)^{a_2}(1+a_3\sqrt{x})} \quad \text{(valence)}. \tag{10}$$

Additionally, the sum of the average momentum fractions of the quarks and gluon must add up to 1 (i.e. 100% of the parent hadron's total momentum) in order for the PDFs to be physically and probabilistically valid. That is,

$$\sum_i \int_0^1 \mathrm{d}x\, x\, f_i(x, Q^2) = 1,$$

where i denotes the ith parton type. However, because we are only considering the u_v, $\bar{u}$, d_v, $\bar{d}$, s, and $\bar{s}$ quarks, the total average momentum fraction of these quarks is less than 1, primarily since we are ignoring the contribution due to the gluon. As such, the total momentum sum rule that will be imposed in this project is

$$\sum_i \int_0^1 \mathrm{d}x\, x\, f_i(x, Q^2) = n, \tag{11}$$

where the average quark momentum fraction $n < 1$ can be estimated to be between 0.5 and 0.6 (i.e. 50% to 60% the total momentum) based on experimental measurements of the gluon PDF $xg(x)$ [9, 10]. For simplicity, if we assume that the antimatter/sea quarks $\bar{u}$ and $\bar{d}$ have the same normalization factor a_0, we can substitute all of the PDFs in our global analysis into this equation and solve for the common sea quark normalization factor:

$$a_0 = \frac{n - a_0^{(u_v)} \int_0^1 \mathrm{d}x\, x u_v - a_0^{(d_v)} \int_0^1 \mathrm{d}x\, x d_v}{(2+\kappa)\left(\int_0^1 \mathrm{d}x\, x(\bar{u}+\bar{d})\right)} \quad \text{(sea)}, \tag{12}$$

where it should be noted that u_v, d_v, $\bar{u}$ and $\bar{d}$ refer to the PDFs with their normalization factor $a_0^{(i)}$ explicitly factored out.

Computational Implementation

Parameter fitting with χ^2 minimization

Given the F_2 structure function in eq. 7 and the general parametrized form of a PDF in eq. 1, the task is to:

1. Guess initial parameters for $u_v(x)$, $\bar{u}(x)$, $d_v(x)$, and $\bar{d}(x)$
2. Calculate a theoretical value for $F_2(x)$ and then compute its χ^2 relative to each data point within each experimental dataset
3. Compute a "global" χ^2_g defined as the sum of the individual χ^2's

4. Apply an optimization algorithm to determine the parameters $\{a_i\}$ that minimize χ^2_g within a given threshold

In the experimental datasets considered in this project, we can assume that the significant errors are uncorrelated so that the χ^2 for the nth experiment/dataset is

$$\chi^2_n(\vec{a}) = \sum_i \frac{\left(F_2(x_i;\vec{a}) - D_n[i]\right)^2}{\sigma^2_{n,\,stat}[i] + \sigma^2_{n,\,sys}[i]}, \tag{13}$$

where $\vec{a}$ is the set of all the parameters used to calculate $F_2(x)$, $D_n[i]$ is the ith data point in the nth experimental dataset, $\sigma_{n,\,stat}[i]$ is the ith standard deviation of the nth experimental dataset (statistical uncertainty), $\sigma_{n,\,sys}[i]$ is the ith systematic uncertainty of the nth experimental dataset, and x_i is the momentum fraction that the ith data point was measured at. Assuming that the data in the experiments generally agree with each other, we can weigh each χ^2 equally so that the global χ^2_g is

$$\chi^2_g(\vec{a}) = \sum_n \chi^2_n(\vec{a}). \tag{14}$$

Gradient Descent

When I started this project, I initially planned to use the gradient descent algorithm (with "momentum" [11]) to minimize the global χ^2_g. However, as I presented in class, the gradient descent algorithm is unable to properly minimize eq. 14 due to the nonlinearity of the F_2 structure function, eq. 7. In the calculations that I ran throughout this project, the gradient descent algorithm diverged if it was left to iterate for too long, underscoring its instability when faced with a nonlinear cost function such as eq. 14. To amend this, I have recently implemented the Levenberg-Marquardt algorithm, which is better suited for least-squares minimization.

Levenberg-Marquardt

The Levenberg-Marquardt algorithm works by interpolating between the Gauss-Newton method and Gradient descent, taking into account the curvature in the parameter space through an approximate Hessian matrix calculated from the Jacobian of the residuals that are to be minimized [12, 13]. Following the cited documentation, I programmed a modified version that takes in additional inputs for parameter space constraints and allows for the parameters to be fixed given a set of conditions (like the sum rules in eq. 10 and eq. 12). The full algorithm can be seen in the accompanying code, so I will not go into detail here. However, I would like to note that I programmed the algorithm with generality in mind so that it can be used for any nonlinear minimization problem beyond what I am considering in this project.

A key property of the LM algorithm is that as it decreases error, the optimization shifts towards the Gauss-Newton method and applies larger changes to the parameters, leading to faster convergence if the algorithm determines the correct direction to optimize in. However, if error increases, the algorithm instead shifts towards gradient descent, applying smaller

changes to the parameters until convergence is reached, or until the steps become too small to move through parameter space. This adaptivity between the Gauss-Newton method and gradient descent makes the LM algorithm robust enough to handle nonlinear optimization problems like the minimization of χ^2_g.

Calculating uncertainty with the Hessian matrix method

After converging on a set of best fit parameters via the LM algorithm, I calculated the uncertainty in the PDF fits using the Hessian matrix method, which has served as the standard approach to uncertainty estimation since the earliest PDF calculations. Under this paradigm, the first step is to calculate the full Hessian matrix H of the global χ^2_g around the best fit parameters $\vec{a}_0$ using finite-difference approximation. The components of H are given by:

$$\begin{aligned} H_{ij} &= \frac{1}{2}\frac{\partial^2 \chi^2_g}{\partial a_i \partial a_j}\bigg|_{\vec{a}=\vec{a}_0} \\ &\approx \frac{\chi^2_g(a_i^0+h, a_j^0+k) - \chi^2_g(a_i^0+h, a_j^0-k) - \chi^2_g(a_i^0-h, a_j^0+k) + \chi^2_g(a_i^0-h, a_j^0-k)}{4hk}, \end{aligned} \tag{15}$$

where h and k are arbitrarily small step parameters and a_i^0 is the ith best fit parameter.

After determining H, the next step is to diagonalize the matrix to find all of its eigenvalues ϵ_k and eigenvectors $\vec{v}$, which can be done numerically via the QR algorithm. Afterwords, we can construct a matrix M whose columns are the eigenvectors $\vec{v}$ scaled by a factor of $T/\sqrt{\epsilon_k}$, i.e.

$$M = \frac{T}{\sqrt{\epsilon_k}}[v^k], \tag{16}$$

where T is an arbitrary tolerance parameter that estimates the uncertainty in χ^2_g. By convention, T is set between 5 and 15 [1, 2]. As explained in [2, 3, 4], the columns of M are the uncertainties $\Delta\vec{a}$ of the best fit parameters $\vec{a}_0$ due to a change in the kth parameter, a_k. That is,

$$\Delta\vec{a}_k = M[:,k] = \frac{T}{\sqrt{\epsilon_k}}v^k, \tag{17}$$

where the notation $M[:,k]$ means to take the element of every row in column k of M.

In order to determine the uncertainty in a fitted PDF, $f(x,\vec{a}_0)$, we can compute its upper and lower bounds due to a change in the kth parameter a_k via

$$f_k^+(x) = f(x,\, \vec{a}_0 + \Delta\vec{a}_k) \tag{18}$$

$$f_k^-(x) = f(x,\, \vec{a}_0 - \Delta\vec{a}_k), \tag{19}$$

which propagates the uncertainty in $\vec{a}_0$ due to a change in a_k to the PDF f. We can then

define the net upper and lower bounds of the uncertainty in f as

$$\Delta f^+(x) = \sqrt{\sum_k \left(f_k^+(x) - f_0(x)\right)^2} \tag{20}$$

$$\Delta f^-(x) = \sqrt{\sum_k \left(f_0(x) - f_k^-(x)\right)^2}, \tag{21}$$

where $f_0(x) = f(x, \vec{a}_0)$ is the central PDF evaluated with the fitted parameters $\vec{a}_0$. By computing $f(x) + \Delta f^+(x)$ and $f(x) - \Delta f^-(x)$, we can finally determine the upper and lower error bands of the PDFs.

Gram-Schmidt QR Decomp. → QR Algorithm → Gauss-Jordan Elimination

Before going into the results of the project, I would like to briefly comment on the eigenvalue/eigenvector pipeline that I implemented to solve for the eigenvalues ϵ_k and eigenvectors $\vec{v}$ of the Hessian matrix H. As we have learned in class, the QR algorithm is the state-of-the-art method for diagonalizing a matrix. In order to perform the diagonalization, one must first decompose the matrix into an orthogonal matrix Q and an upper triangular matrix R, which I have chosen to do via the classical Gram-Schmidt QR decomposition. After obtaining the diagonal matrix of eigenvalues, one can then solve for the corresponding eigenvectors, which I implemented through the standard Gauss-Jordan elimination procedure.

However, it should be emphasized that neither the classical Gram-Schmimdt decomposition nor the Gauss-Jordan elimination techniques are optimal for decomposing and solving the Hessian of the global χ^2_g, since – again – χ^2_g is nonlinear with respect to the parameters. As a result, H acquires a wide range of values spanning multiple orders of magnitude, creating numerical noise and instability within both of the aforementioned algorithms. Although I have programmed these algorithms and tested them on smaller, well behaved matrices, I ultimately decided to discard my eigenvalue/eigenvector pipeline in favor of NumPy's "eig" method, which performs the same steps but with more sophisticated QR decomposition routines and eigenvector solvers [14].

Results

For my simplified global analysis, I only considered data from *ep* DIS experiments that measured the F_2 structure function at an average energy scale of $Q^2 = 15$ GeV. The experiments were performed by the H1 and ZEUS collaborations at HERA and are discussed further in their corresponding papers [15, 10, 16, 17, 9, 18]. Combined, these data sets contribute 44 points to constrain 12 parameters (3 each for u_v, $\bar{u}$, d_v and $\bar{d}$ since the normalization factor a_0 is fixed by the sum rules in eq. 10 and eq. 12), leaving a total of 32 degrees of freedom to fit with.

In order to optimize the χ^2_g of the PDFs, I initialized the LM algorithm with a set of arbitrarily chosen initial parameters that constrained the quark momentum fraction to approximately 50% and closely matched the trend of the data, but did not necessarily minimize

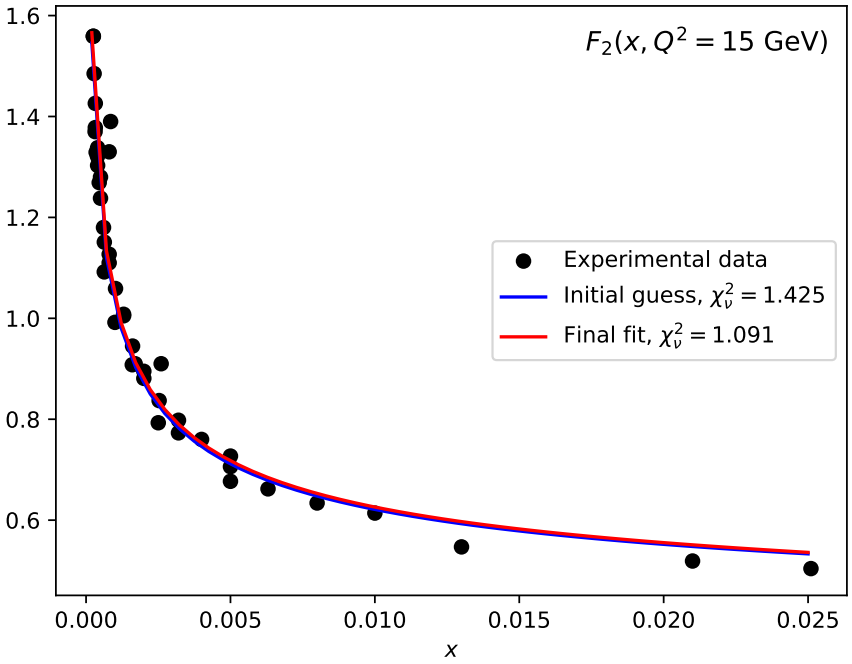


Figure 1: Fitted F_2 Structure Function at fixed $Q^2 = 15$ GeV. Both the initial guess (blue) and final fit (red) are shown.

the χ^2_g. Since the parameter space of χ^2_g can potentially have multiple minima, it was essential that the initial parameters were close to the expected fit, or else the algorithm could have converged on a completely different, unphysical minima that did not properly fit the data. In order to determine the "goodness" of the fit, I calculated the reduced χ^2 of the global χ^2_g, which is given by

$$\chi^2_\nu = \frac{\chi^2_g}{\nu}, \tag{22}$$

where ν is the number of degrees of freedom ($\nu = 32$ in this case). In general, a statistically acceptable χ^2_ν is around $\chi^2_\nu \approx 1 \pm 0.1$, where a reduced value $\gg 1$ indicates underfitting while a value $\ll 1$ indicates overfitting. As shown in fig. 1, the initial parameters that I inputted resulted in a reduced χ^2_g of $\chi^2_\nu \approx 1.425$, which slightly underfit the trend of the data. After optimization with the LM algorithm, the parameters were adjusted such that the final fitted reduced χ^2_g became $\chi^2_\nu \approx 1.091$, which is within the statistically acceptable range.

After converging on a set of parameters that minimized the reduced χ^2_ν to within $\approx 1 \pm 0.1$, I performed the uncertainty calculations outlined in the previous section and obtained the following finalized plot for the parton distribution functions of the u_v, $\bar{u}$, d_v, $\bar{d}$, and $s = \bar{s}$ quarks, as shown in fig. 2.

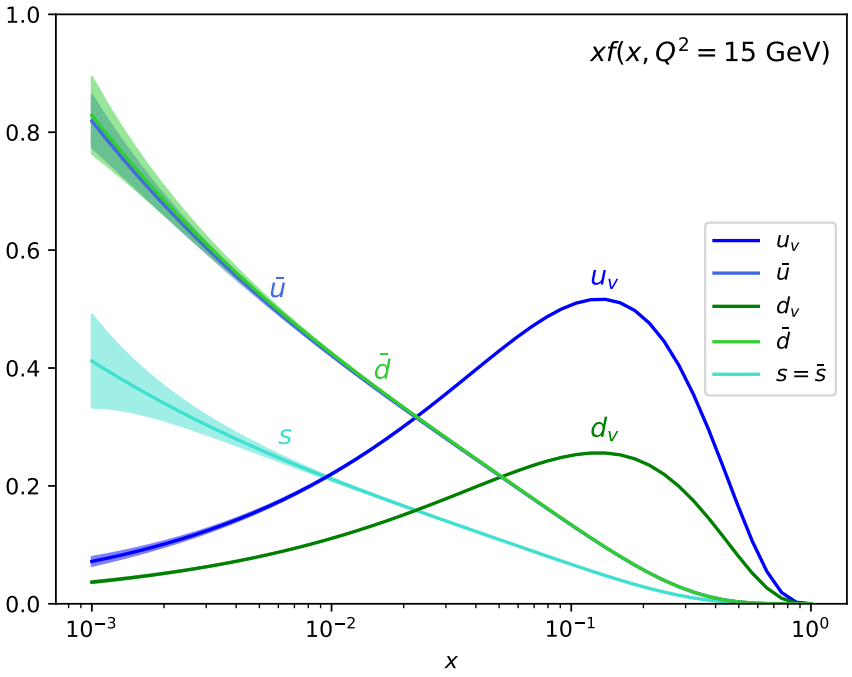


Figure 2: Fitted PDFs at fixed $Q^2 = 15$ GeV for u_v, $\bar{u}$, d_v, $\bar{d}$, and $s = \bar{s}$ with estimated uncertainty bands.

Discussion

Qualitatively, the fitted PDFs in fig. 2 are similar to the fits produced by the leading global analysis collaborations, namely NNPDF, MSHT, and CTEQ-TEA. Similar to their findings, my fits for the u_v and d_v quarks peak around the higher momentum fraction regime, while the sea quarks $\bar{u}$, $\bar{d}$, s, and $\bar{s}$ dominate the lower momentum fraction regime. This particular distribution indicates, for example, that if a parton were found with a relatively high momentum fraction, it is most likely to be a u_v quark, second most likely to be a d_v quark, and almost never a sea quark.

However, unlike the plots produced by the leading PDF groups, my plot severely underestimates the uncertainty in the middle-to-higher momentum fractions. Typically, uncertainty increases around the peaks of the u_v and d_v distributions due to various sources of systematic error when measuring partons with higher momentum and energy. It is likely that these errors compounded upon each other, leading to correlated uncertainties that I did not account for in my error estimation. As a result, the uncertainty bands shown in fig. 2 appear suppressed and do not provide an accurate representation of the true uncertainty in each of the fitted PDFs.

It should also be noted that the plot shown in this paper is slightly different from the plot that I presented in class, which was made before I reworked the optimization scheme into the LM algorithm. In the figure that I previously obtained using the gradient descent method, the $\bar{u}$ and $\bar{d}$ PDFs deviated from each other near the lower momentum fraction regime, which does not particularly agree with the PDFs produced by the aforementioned global analysis

groups. Now, with the parameters set by the LM algorithm, the $\bar{u}$ and $\bar{d}$ PDFs generally coincide with each other near low x, more closely matching the plots established in the literature. However, due to sea quark asymmetry, it should be expected that the $\bar{d}$ quarks peak around $x = 10^{-1}$, which is markedly not present in fig. 2. Due to the parameterized form that I have chosen and the energy scale that I am considering, this small peak was overlooked in favor of fitting the general trend of the PDFs.

Conclusion

Global analysis of parton distribution functions is a notoriously difficult task, requiring state-of-the-art algorithms that improve upon the classical methods that we have seen in class. Although I initially attempted to perform a simplified global analysis using gradient descent and an eigenvalue/eigenvector solver based on classical Gram-Schmidt QR decomposition and Gauss-Jordan elimination, I ultimately decided that they were insufficient for accurately fitting PDFs and determining their uncertainties. As such, I reworked the workflow of my code to incorporate the Levenberg-Marquardt algorithm and NumPy's built in eigenvalue/eigenvector solver, allowing me to compute qualitatively reasonable PDFs for the u_v, $\bar{u}$, d_v, $\bar{d}$, and $s = \bar{s}$ quarks.

Moving forward, it would be imperative to incorporate the DGLAP equations into the global analysis, as they provide a method to evolve the PDFs into other energy scales Q^2. Such flexibility would allow one to consider a much larger set of data as well as determine constraints on the gluon's PDF, which was emphatically ignored in this simplified analysis. Additionally, one must also consider data from Drell-Yan processes and jet production measurements, which provide further constraints on the PDFs and open up the possibility to account for heavier quarks like the charm.

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